Recall: • abelian categories $\mathrm{DMod}^l(X) \underset{-\otimes_{\mathcal{O}_X}\omega_X^{-1}}{\overset{-\otimes_{\mathcal{O}_X}\omega_X}{\rightleftarrows}} \mathrm{DMod}^r(X)$

$f_*(M) = f_*^{\mathcal{O}}(M \otimes_{D_X} \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y)$

• $f: X \to Y \rightsquigarrow f^+ : \mathrm{DMod}^l(Y) \to \mathrm{DMod}^l(X)$

$f_* : \mathrm{DMod}^r(X) \to \mathrm{DMod}^r(Y)$

$T^*X \leftarrow T^*Y \times_Y X \to T^*Y$

$\mathcal{O}_X \otimes_{\mathcal{O}_Y} D_Y$

compute using transfer bimodules: $D_{X\to Y} = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y$

use this to express $f_*$ on left D-mod $\longrightarrow$ $D_{Y\leftarrow X} = \omega_X \otimes_{\mathcal{O}_X} D_{X\to Y} \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\omega_Y$ [check notes for formula]

$D_X$ $D_Y$

$= \omega_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y \otimes_{f^{-1}\omega_Y} f^{-1}\omega_Y$

• $\mathrm{DMod}_c(X)$ coherent (i.e. f.g.) D-modules (i.e. over $D_X$)

$\mathrm{DMod}_h(X)$ holonomic D-modules (minimal singular supp)

finite length

<u>flat connection</u> = D-mod, where underlying $\mathcal{O}$-mod is finite rank, loc. free.

<u>Goal</u>: Understand how holonomic D-modules "look constructible"

<u>Def</u>: $D^+(\mathrm{DMod}\,X)$ $D^-(\sim)$ $D^b(\sim)$ $D(\sim)$

all gradings cohomological ... $d: M^0 \to M^1$

complexes with coherent cohomology

complex of coherent D-mod

$D(\mathrm{Mod}_c(X)) \xrightarrow{\simeq} D_c(\mathrm{DMod}\,X)$

$D(\mathrm{Mod}_h(X)) \longrightarrow D_h(\mathrm{DMod}\,X)$

<u>Def</u>: $\dim(f: X \to Y) = \dim(X) - \dim(Y)$

underlying $\mathcal{O}$-mod functor is $f^*$

maybe little mixes L, R.

$f^! : D(\mathrm{DMod}\,Y) \to D(\mathrm{DMod}\,X)$

$f_* : D(\mathrm{DMod}^r X) \to D(\mathrm{DMod}^r Y)$

$f^!M = Lf^+M[\dim f]$

$f_*^{\text{right}}M = Rf_*^{\mathcal{O}}(M \otimes^L_{D_X} \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y)$

$f_*^{\text{left}}M = Rf_*(\omega_X \otimes^L_{D_X} M \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}D_Y \otimes_{f^{-1}\omega_Y} f^{-1}\omega_Y)$

Ex. $X \longrightarrow * \longleftarrow X$

$\mathcal{O} \leftsquigarrow k[-\dim X] \leftsquigarrow \mathcal{O}$

$\mathcal{O} \otimes \mathcal{M}_X \leftsquigarrow \mathcal{M}_X[-\dim X] \leftsquigarrow \mathcal{M}$

Ex.: $\{0\} \xhookrightarrow{f} \mathbb{A}^1$ $f$ proper

Fact: $(f_*, f^!)$ expect $(f^*, f_*)$ ⇝ defect 1

$R\mathcal{H}om(k, f^! \mathcal{M}) = f^! \mathcal{M} = k \overset{L}{\otimes}_{\mathcal{O}_X} \mathcal{M}[-1] =$ $\mathcal{M}$ ⇝ $H^1$ cokern; $\uparrow x$; $\mathcal{M}$ ⇝ $H^0$ kernel

$R\mathcal{H}om(f_* k, \mathcal{M}) = R\mathcal{H}om(\mathcal{D}/\mathcal{D}x, \mathcal{M}) =$ $\mathcal{M}$; $\uparrow x$; $\mathcal{M}$ ⇝ $H^0$ kernel

$= \mathcal{H}om(\mathcal{D}, \mathcal{M}) =$ ($\uparrow x$, $\mathcal{D}$)

Ex.: $f: X \longrightarrow *$

Spencer resolution: deformation of $dR_X \otimes_{\mathcal{O}_X} \mathcal{D}_X$

$$0 \to \Omega^0_X \otimes_{\mathcal{O}_X} \mathcal{D}_X \to \Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{D}_X \to \cdots \to \Omega^n_X \otimes_{\mathcal{O}_X} \mathcal{D}_X \to \omega_X = \Omega^n_X \to 0$$

$dR^\bullet(\mathcal{D}_X)$

resolution, locally free

$$\omega \otimes P \longmapsto d\omega \otimes P + \sum \underbrace{(dx_i \wedge \omega) \otimes \frac{d}{dx_i} P}_{\text{"deformation term"}}$$

local

Exercise: Check this is a resolution

Hint: Write filtration of complex, gradded, get Koszul complex (acyclic)

Def: $dR(\mathcal{M}^\bullet) = dR^\bullet(\mathcal{D}_X) \otimes_{\mathcal{D}_X} \mathcal{M}$. "de Rham complex of $\mathcal{M}$"

left D-module

$f_*(\mathcal{M}) = R\Gamma(X, dR(\mathcal{M}^\bullet))$

Remark $f_*$ in general is hard

Gauss-Manin connection

$\Rightarrow f_*(\mathcal{O}_X) = H^\bullet_{dR}(X)$

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Goal: Thm (preservation): $f^!$, $f_*$ preserve holonomic D-mod.

Thm (constructibility): A complex $\mathcal{M}^\bullet$ of D-mod. is holonomic iff there is a stratification $S_\alpha \xhookrightarrow{i_\alpha} X$ such that $i^!_\alpha \mathcal{M}$ are flat connections,

Warning: $f^*$ on D-modules is not generally $Lf^* \underset{\mathcal{O}}{=} Lf^{\mathcal{O},*}$

Temporary def$^n$: $\mathcal{M}^\vee = R\mathcal{H}om(\mathcal{M}, \mathcal{D}_X)$ $\qquad$ $\mathcal{M}$ left $\rightsquigarrow$ $\mathcal{M}^\vee$ right

degrees $[0, \infty)$

Thm (Roos) $\mathcal{M} \in D\mathrm{Mod}_c(X)$ chart. Then

① $\dim(SS(\mathcal{H}^k(\mathcal{M}^\vee))) \leq 2\dim(X) - k$

② If $\operatorname{codim}(SS(\mathcal{M})) = k$, then

$\mathcal{M}^\vee$ has coh. amplitude $[k, n]$ ← charity shows is degree $k, \dots, n$

$\mathcal{M}^\vee =$

| | |
|---|---|
| $n$ | $\leq n$ |
| | dim upper bd |
| $2$ | $\vdots$ |
| $1$ | $\leq 2n-1$ |
| $0$ | $\leq 2n$ |

One consequence: If $\mathcal{M}$ holonomic, then by ②, $k = n$

$\mathcal{M}^\vee$ is in degree $n$.

Def$^n$: $\mathbb{D}_X(\mathcal{M}) = \mathcal{E}^{-1}(R\mathcal{H}om(\mathcal{M}, \mathcal{D}_X)[\dim X])$ ← $\mathbb{D}_X : D\mathrm{Mod}_h(X) \to D\mathrm{Mod}_h(X)^{op}$

site charity

- exact
- equivalence

Prop: $\mathbb{D}_X : D_c D\mathrm{Mod}(X) \to D_c D\mathrm{Mod}(X)^{op}$

- $\mathbb{D}_X^2 = id$
- $\mathbb{D}_X : D_c^- D\mathrm{Mod}(X) \xrightarrow{\simeq} D_c^+ D\mathrm{Mod}\, X^{op}$
- $\mathbb{D}_X : D\mathrm{Mod}_h(X) \xrightarrow{\simeq} D\mathrm{Mod}_h(X)^{op}$

use them to define

- $f_! = \mathbb{D} f_* \mathbb{D}$
- $f^* = \mathbb{D} f^! \mathbb{D}$ ← chart

might not preserve.

$D^b_{rh}(X) \xrightarrow{\simeq} D^b_c(X)$

$\cup \qquad\qquad \cup$

$D\mathrm{Mod}_{rh}(X) \xrightarrow{\simeq} \mathrm{Perv}(X)$

std ⟷ perv.

?? ⟷ std

t-str. $D^b_{rh}(X)$ extend to $D^b(X)$ ??

Question ??

What is non-std t-str. on $D^b_m$